

PARAMESHWARAN ANANDHAKUMAR

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SUMMARY

Software Engineer with 4+ years building **backend services**, **distributed data pipelines**, and **production ML systems**. At Ford, owned the **Java/Spring Boot REST API** platform behind **500K+ monthly transactions** across 11 dealer networks, along with the **Apache Spark**, **Airflow**, and **dbt** pipelines and **Snowflake** models feeding it and the **Jenkins**, **GitLab CI** automation that shipped it. Depth across **Python/FastAPI microservices**, **model serving** and **inference optimization**, **AWS (SageMaker, Lambda, EC2, S3)**, and **Docker/Kubernetes**.

SKILLS

Languages: Python, Java, SQL, Scala, TypeScript, JavaScript, C++, Bash, PySpark, Spark SQL

Backend & APIs: FastAPI, Spring Boot, Django, Node.js, Express, REST API design, GraphQL, microservices, event-driven architecture, Swagger/OpenAPI, React.js

Big Data & Pipelines: Apache Spark, Apache Airflow, dbt Core, Kafka, ETL/ELT pipeline design, data ingestion & validation frameworks, data pipeline architecture

ML & AI Systems: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Sentence Transformers, feature engineering, model training & evaluation, model serving, inference optimization, RAG, FAISS, LangChain

MLOps & DevOps: Docker, Docker Compose, Kubernetes, Jenkins, GitLab CI, GitHub Actions, AWS CodePipeline, CI/CD automation, Nginx, Prometheus, Grafana

Cloud & Warehousing: AWS (S3, EC2, Lambda, SageMaker, EKS, Glue), Snowflake, PostgreSQL, MySQL, MongoDB, Redis, schema design, indexing strategy, partitioning, EXPLAIN plan analysis

Engineering Practices: Agile/Scrum, Git, system design, code & design review, pytest, JUnit, Jest, incident triage, root cause analysis, technical documentation

EXPERIENCE

Software Engineer | University at Buffalo, Institute for AI and Data Science

Jun 2026 – Present

- Cut an external **LLM API** out of the critical path of UB Atlas, a research-to-funding matching platform, replacing it with **self-hosted Sentence Transformer embeddings** (all-MiniLM-L6-v2) and **cosine similarity** ranking over 1,821 faculty profiles, taking per-query API cost to zero and search latency **under 100ms**.
- Shipped the platform end to end: a **Node.js/Express REST API** documented in **Swagger/OpenAPI**, a **Python ML matching service**, **MongoDB** persistence, and a **React 18/TypeScript** front end with instant command-palette search.
- Deployment had required a multi-terminal setup, so **containerized** the stack behind **Docker Compose** and an **Nginx** reverse proxy with health checks and environment-scoped configs, reducing it to one reproducible command.
- Engineered a record reconciliation **data pipeline** that normalized identifiers across 275 scraped affiliate records and a 1,821-document store, reaching an 83% match rate with zero duplicates and covered by an automated **API**, **ML**, and **database** test suite.

Software Engineer | Ford Motor Company

Jul 2022 – Jul 2025

- Designed and shipped the **REST API** platform layer for order management, pricing, and delivery on the Service Parts After Sales system in **Java**, **Spring Boot**, **MySQL**, and **React.js**, sustaining **500K+ transactions a month** across 11 dealer networks and cutting order processing **55%**.
- Dealer inventory and parts demand data moved through **distributed pipelines** built on **Apache Spark**, **Apache Airflow**, and **dbt Core**, modeled in **Snowflake** and cut to **20%** lower **ETL** latency, giving downstream teams one consistent view across 11 networks.
- Productionized a **demand forecasting** and vendor priority scoring model as a real-time **inference service** in **Python**, **FastAPI**, and **PostgreSQL**, serving **time series predictions** that reduced stockout risk **38%** across 3 networks.
- Optimized **MySQL** warehouse performance across 2M+ daily records by rewriting 50+ queries against **EXPLAIN** plans, reworking **schemas** and **indexing strategy**, and adding table **partitioning**, cutting average query runtime **70%** while holding **throughput** stable at peak load.
- Replaced a largely manual release process with automated **CI/CD** across **Jenkins**, **GitLab CI**, and **AWS CodePipeline**, adding **Docker**-based build and test orchestration that shortened release cycles **30%**.
- Owned these services end to end across **design review**, **test coverage** (**JUnit**, **pytest**), deployment, and **production support**, working with globally distributed engineering and dealer operations teams and converting **incident triage** and **root cause analysis** into permanent fixes.

AI/ML Software Engineer | RMD Engineering College

Jun 2019 – Sep 2020

- Trained and benchmarked custom **CNN**, **AlexNet**, and **LeNet** architectures for **MRI classification** to 92% validation accuracy, a 20% gain over baseline, on **AWS SageMaker** with **S3**-backed dataset versioning and **EC2** training instances.
- Deployed the selected model behind a **Django REST** service fronted by **AWS Lambda**, returning **inference** in under 1 second and scaling to demand without idle compute.

AI SYSTEMS PROJECT

AgentServe | Agent-Aware LLM Serving Gateway for Kubernetes GPU Clusters

Built an **LLM serving gateway** in **Python** and **FastAPI** that schedules multi-turn **agent sessions** rather than single requests across **vLLM** replicas on **Kubernetes**, pairing **session affinity load balancing** with an online **EWMA** predictor of tool call duration and a three tier **KV cache** policy that offloads paused sessions to **host DRAM** instead of evicting them; lifted **cache hit rate** from 33% to 71% and cut **GPU compute time 81%** on replayed coding agent traces, shipped with **Redis**-backed distributed routing state, **SSE streaming**, **Docker** and **Helm** deployment, **Prometheus** observability and 135 tests across 6 **GitHub Actions CI** jobs.

EDUCATION

Master of Science, Data Science | University at Buffalo, New York

BTech, Information Technology | Anna University, Chennai, India